

# Adaptive Federated Optimization via Dynamic Gradient Quantization for Heterogeneous Edge Networks

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**Abstract**—Federated Learning (FL) enables geographically distributed edge devices to collaboratively train machine learning models while retaining raw training data locally. However, communication overhead remains a primary bottleneck when deploying FL models over low-bandwidth, unreliable wireless edge networks. In this paper, we propose DYNQUANT-FL, a dynamic gradient quantization framework tailored for resource-constrained edge computing environments. By dynamically adjusting quantization precision on a per-layer basis according to local gradient variance, network latency metrics, and available channel capacity, DYNQUANT-FL achieves up to an 84.2% reduction in total uplink data transfer without sacrificing final model accuracy. We validate our framework across a 32-node hardware testbed comprising Nexa Edge-V1 modules and Apex embedded units under non-IID data distributions and fluctuating channel profiles. Experimental results demonstrate that DYNQUANT-FL accelerates global convergence by  $2.35\times$  compared to standard FedAvg and outperforms static quantization baselines by up to 14.8% in top-1 accuracy under severe bandwidth constraints ( $< 1$  Mbps).

**Index Terms**—Federated learning, gradient quantization, edge computing, distributed optimization, adaptive compression, heterogeneous networks.

## I. INTRODUCTION

The proliferation of Internet of Things (IoT) sensors, autonomous robotics, and mobile devices has led to exponential growth in telemetry data generated at the network edge. Traditional centralized machine learning paradigms require transferring massive datasets from edge nodes to centralized cloud datacenters. This centralized model poses severe scalability challenges due to backhaul network congestion, strict data privacy regulations, and unacceptable communication latencies [1], [2].

Federated Learning (FL) has emerged as a compelling privacy-preserving paradigm by shifting model training from central servers to distributed edge clients [3]. In FL, edge devices compute local model updates on their native datasets and periodically transmit parameter vectors to a central

coordinator. The server aggregates these updates—typically via Federated Averaging (FedAvg)—to produce a refined global model, which is subsequently redistributed to the participating clients.

Despite its advantages, deploying FL in real-world wireless edge networks encounters fundamental operational obstacles:

- 1) **High Communication Overhead:** Deep neural network models frequently contain tens to hundreds of millions of parameters. Transmitting full-precision 32-bit floating-point ( $fp32$ ) gradient updates across hundreds of communication rounds severely degrades throughput on bandwidth-constrained links.
- 2) **System and Channel Heterogeneity:** Edge nodes possess highly diverse compute capabilities, memory limits, and uplink bandwidth profiles. Devices on congested cellular networks or low-power wireless links act as stragglers, delaying global synchronization rounds.
- 3) **Statistical Data Heterogeneity:** Real-world edge data distributions are inherently Non-Independent and Identically Distributed (non-IID). Statistical skew causes local gradient vectors to diverge substantially across clients, rendering static quantization schemes prone to severe error accumulation and training instability.

To address these interconnected challenges, we propose DYNQUANT-FL (Dynamic Quantization for Federated Learning), an adaptive compression framework designed specifically for heterogeneous edge networks. Unlike conventional quantization approaches that impose static bit-widths across all network layers and training epochs, DYNQUANT-FL continuously monitors empirical gradient variance and channel state information (CSI) to allocate optimal quantization precision dynamically.

### A. Key Contributions

The main contributions of this paper are summarized as follows:

- **Adaptive Rate-Distortion Formulation:** We model gradient compression as a joint optimization problem balancing

This research was supported in part by the Meridian Science Foundation under Grant No. FX-2025-8810, the Apex Autonomous Systems Research Initiative, and Vertex Systems Lab.